

JAINENDRA KUMAR (JK) SINGH

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TECHNOLOGY LEADER | ENTERPRISE PLATFORMS & CLOUD | GENAI & AI STRATEGY

- Fractional CTO and technology executive with 30 years of experience driving enterprise-scale cloud, data, and GenAI transformation for Fortune 500 and growth-stage organizations across regulated and complex industries.
- Strategic advisor to C-suite leaders on Cloud transformation, Application & platform modernization, and GenAI driven transformation across diverse industries. Built multiple COEs and architecture frameworks that accelerate large scale technology modernization. Extensive experience in managing executive stakeholder alignment, shaping MVP roadmaps, overseeing POC execution, and driving GenAI-based innovation for Fortune 500 enterprises and Startups.

CORE COMPETENCIES

- **GenAI & Innovation:** Agentic AI, GenAI/RAG Architectures, Model Context Protocol (MCP), AI Platform Strategy, LLM Fine-tuning & Optimization, AI Safety & Governance, Product Innovation
- **Strategic Leadership:** P&L Management (>\$50M accounts), C-Suite Advisory, Strategic Planning, Business Development, RFP Leadership, Executive Workshops, Global Team Leadership (1,000+ professionals), Stakeholder Management, Budget & FinOps
- **Cloud & Architecture:** Multi-Cloud Strategy (AWS, Azure, GCP), Cloud Governance & Security, Enterprise & Solution Architecture, Cloud Migration & Modernization, Legacy/Mainframe Transformation, Hybrid Cloud Design
- **Platform & Delivery:** DevOps Strategy, CI/CD Pipelines, Containerization (Kubernetes, OpenShift), CRM & Customer-Facing Platforms, Platform Engineering, Architecture Governance, Centers of Excellence, Data Platforms

PROFESSIONAL EXPERIENCE

HyLifen Chantilly, VA

Jul 2024 – Present

Fractional CTO & Strategic Technology Advisor

Fractional CTO for multiple organizations, providing strategic technology leadership that accelerates MVP development, de-risks investments, and executes enterprise-scale cloud and GenAI transformation.

- Currently leading implementation of AI assisted development framework for a 3 tier Application development environment of a US Government Agency.
- Partnered with C-suite leadership to define technology vision, innovation roadmaps, and competitive differentiation strategies
- Lead executive presentations and strategic RFP response for Fortune 500 clients, Guide MVP development and POC validation for startups
- Leading implementation of AI assisted development framework for a 3 tier Application of a US Government Agency.
- Built GenAI products: MCP servers for intelligent job extraction and matching; crowdsourcing-based Agentic AI solutions
- Healthcare startup: Architected GenAI-based MVP platform with Agentic AI for personalized patient engagement, AI-driven data protection, and specialized patient guidance
- Financial Services: Designed GenAI/RAG solutions generating actionable insights from past records and code bases residing on on-prem, cloud, and mainframe environments for intelligent incident resolution
- Healthcare Payer: Led container platform modernization from Pivotal Cloud Foundry to Red Hat OpenShift

Cognizant Technology Solutions, College Station, TX

Dec 2020 – Jun 2024

Director – Chief Cloud Architect

Spearheaded enterprise-scale AWS cloud architecture and application modernization initiatives while driving business growth and presales strategy across North American accounts. Served as Chief Cloud Architect responsible for solution architecture, cloud Centers of Excellence, modernization frameworks, and CXO-level advisory, supporting large-scale transformations, M&A programs, and regulated industry customers.

- Established and scaled Centers of Excellence across database migration, FinOps, mainframe-to-cloud migration, containerization, and service mesh, standardizing AWS-based platforms and accelerating enterprise adoption.
- Partnered with client sales, operations, and IT leaders to design cloud architectures for customer-facing platforms and core enterprise systems supporting day-to-day business operations.

Cognizant Technology Solutions *Continued*.....

- Led delivery of 120+ high-stakes RFPs across cloud migration, cloud operations, FinOps, and consulting engagements, supporting large-scale enterprise transformation initiatives.
- Generated \$25M in new business in six months and increased the AWS opportunity pipeline by 50% across 75 North American accounts.
- Architected AWS-based solutions for 100+ critical applications as part of a major M&A program for a leading insurance provider, delivering scalable, secure, and operationally resilient cloud architectures.
- Engineered cloud solutions to modernize legacy financial applications for a Fortune 100 company, enabling cloud-native adoption and improved performance.
- Defined and implemented AWS multi-account governance using AWS Organizations, Control Tower, and Service Control Policies (SCPs), and standardized infrastructure provisioning using Terraform and CloudFormation across enterprise cloud platforms.
- Led GenAI initiatives to build enterprise capabilities for rapid application development, including LLM-driven automation, analytics, and development acceleration.

Tata Consultancy Services (TCS), Edison, NJ

Nov 2002 – Dec 2020

Global Technology & Commercial Leadership | Cloud, Data & Enterprise Transformation

Held progressive senior leadership roles over an 18-year tenure with end-to-end responsibility for enterprise technology platforms, global delivery operations, and large-scale transformation programs. Led customer-facing platforms, data systems, CRM, and shared services across regulated industries while building and operating offshore, nearshore, and onshore delivery models.

- Served as executive delivery owner, client partner, and account head for strategic enterprise accounts including World Bank Group, Dow Chemical, Cardinal Health, Nielsen, Walgreens, Genworth, AXA, and CVS Caremark, acting as the single escalation point for delivery, commercial, and contractual matters.
- Directed large-scale transformation programs including CRM modernization, enterprise data platform upgrades, cloud migration factories, and shared services modernization.
- Built and scaled global delivery organizations of 500–1,000+ professionals across India, Southeast Asia, LATAM, Europe, and the United States, with ownership of governance, cost, quality, and delivery outcomes.
- Led modernization of customer-facing and core enterprise platforms including cloud, CRM, data platforms, infrastructure, and shared services.
- Defined and operationalized global operating models, governance structures, transition playbooks, and escalation mechanisms enabling consistent multi-region delivery.
- Chaired executive steering committees with CIOs, VPs, and Directors, driving platform prioritization, transformation roadmaps, and performance governance.
- Partnered with global presales and leadership teams to shape transformation programs, managed services proposals, RFP responses, and long-term account growth initiatives.
- Drove operational excellence and continuous improvement initiatives by implementing standardized metrics, automation opportunities, and delivery best practices to improve quality, efficiency, and predictability across delivery centers and accounts.

ADDITIONAL PROFESSIONAL EXPERIENCE

Aug 1995 – Nov 2002

- Syntel, Société Generale, HCL Perot, Programmer Analyst | Project Manager | Other Roles

TECHNICAL PROFICIENCIES

- **GenAI Platforms & Frameworks:** Agentic AI | Model Context Protocol (MCP) | Agent-to-Agent (A2A) | GPT-5.2 | Claude 4.5 | Gemini 3 | Llama 4 | Qwen3 | Kimi-K2 | OpenAI | Amazon Bedrock | Hugging Face | LangChain | LangGraph | Ollama | CrewAI | vLLM
- **AI Development Tools:** Claude Code | VS Code | Cursor | Codex | AWS Q | AntiGravity | Gradio | Streamlit | FastAPI | FastMCP
- **AWS AI & Data:** Bedrock | Knowledge Bases | SageMaker | Textract | ECS | Lambda | Lake Formation | Redshift | DynamoDB | Aurora | EventBridge | Elasticsearch | Macie | ElastiCache | Glue | Kinesis | IoT Core
- **Azure AI & Data:** AI Foundry | ML Studio | AutoML | Pipelines | Functions | Logic Apps | AI Search | Document Intelligence | Purview | Azure Data Factory | Databricks | ADLS | Synapse | Monitor | Application Insights | Activity Log
- **GCP AI & Data:** Vertex AI | AutoML | GKE | Cloud Workflows | Cloud Functions | Document AI | Memorystore for Redis | Dataplex | Data Fusion | Dataproc with Databricks | BigQuery | Cloud Trace | Cloud Audit Logs
- **AI Safety & Governance:** OpenTelemetry (OTel) | NeMo Guardrails | Topic Control | Jailbreak Detection | ISO 42001 | Secure Multi-Party Computation (SMPC) | SecLM | Moderation APIs | Red Teaming
- **ML & AI Modeling:** Linear/Logistic Regression | Naive Bayes | Decision Trees | Random Cut Forest | Gradient Boosting | XGBoost | NLTK | spaCy | PCA | NLP/NLU | SVM | Transformers | Lightning | Diffusers | Computer Vision | Torchvision
- **Model Training & Evaluation:** Phoenix | LLM-as-a-Judge | GRPO | DPO | Ragas | ROUGE | BLEU | LIME | SHAP | HumanEval
- **Model Optimization:** LoRA | QLoRA | Quantization | Distillation | Adaptive Sparsity | AutoRound | Parameter-Efficient Fine-Tuning
- **Containers & Platform Engineering:** Docker | Kubernetes | OpenShift | Amazon EKS | Jenkins | GitLab | Terraform | CloudFormation | CI/CD Pipelines
- **Programming Languages:** Python | JavaScript | Node.js | SQL | Java | C/C++
- **Databases:** Oracle | SQL Server | DB2 | Sybase | MySQL | PostgreSQL | DynamoDB | MongoDB
- **Operating Systems:** Windows | Linux | Unix

EDUCATION, PROFESSIONAL DEVELOPMENT, CERTIFICATIONS

- Bachelor of Engineering (BE), Electrical & Electronics, National Institute of Technology Karnataka, India
- Chief Technology Officer (CTO) Program @The Wharton School of the University of Pennsylvania
- Scaling a Business – How to Build a Unicorn @The Wharton School of the University of Pennsylvania
- Executive Presence and Influence @The Wharton School of the University of Pennsylvania
- AWS Solution Architect Associate – (2019)
- Healthcare Delivery Providers (From University of Minnesota on Coursera)
- Healthcare Marketplace (From University of Minnesota on Coursera)
- Generative AI Fundamentals from Databricks
- AI Security and Governance Certified – (by security education)
- Oracle Cloud Infrastructure 2025 AI Foundations Associate
- Career Essentials in Generative AI by Microsoft and LinkedIn
- Dell GenAI Foundations
- AI Agents Fundamentals Certification by Hugging Face
- Model Context Protocol- Advanced Topics Certifications by Anthropic
- MCP Certification by Hugging Face